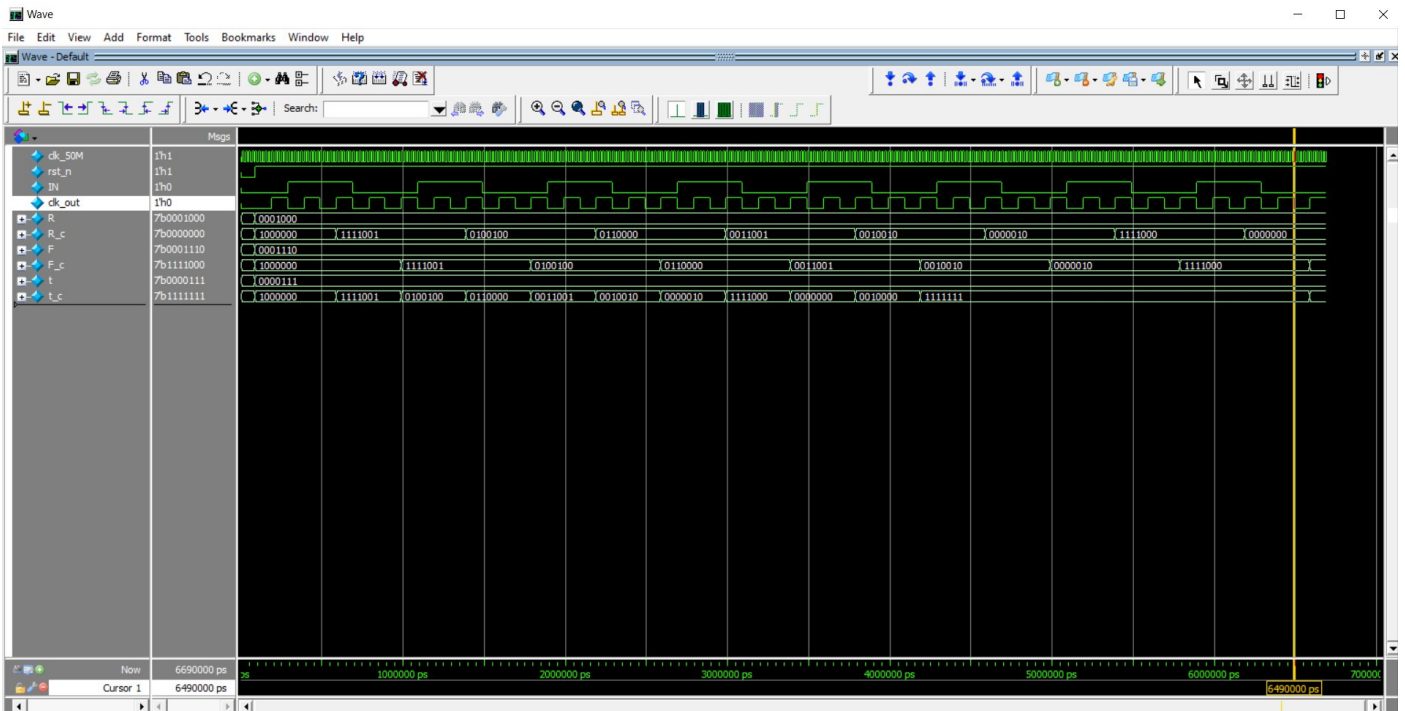
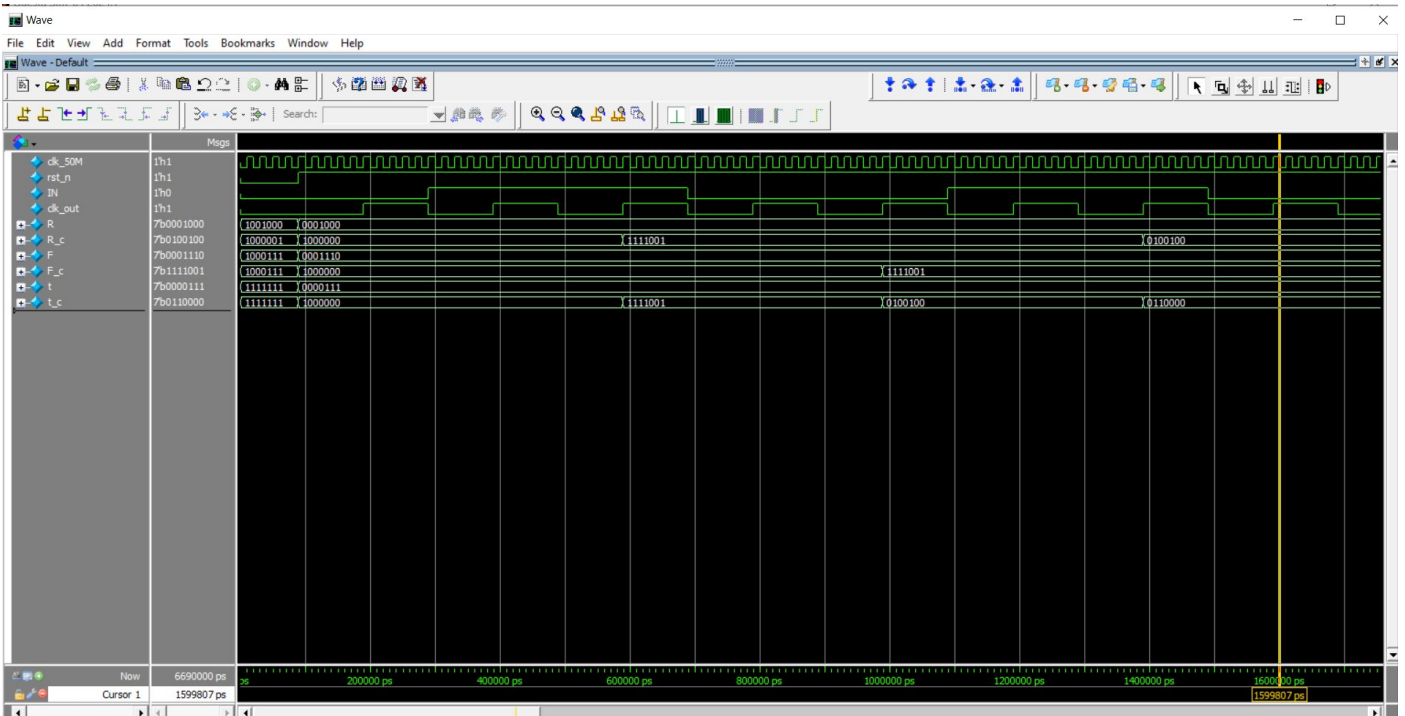


Edge detector

Wave form snapshot:

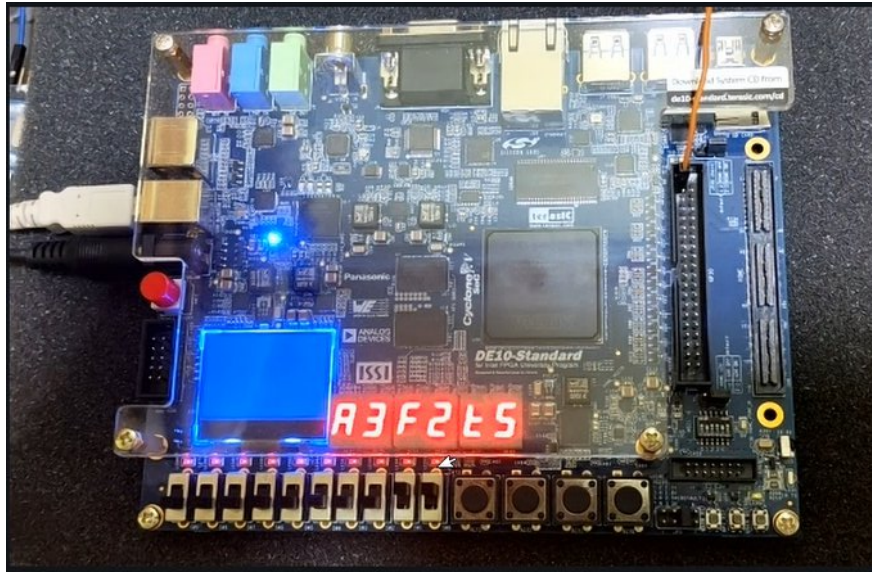


A more zoomed one:



Hardware Implementation & Validation:

Real-Time Operation & Counting



Upon releasing the reset switch ($\text{rst_n} = 1$), the system switches from display override mode to active tracking mode:

- Displays 1 & 2 output the rising edge marker **R** alongside the running rising edge count (**R_c**).
- Displays 3 & 4 output the falling edge marker **F** alongside the running falling edge count (**F_c**).
- Displays 5 & 6 output the total edge marker **t** alongside the combined edge count (**t_c**).

Reset State Verification



When the global active-low reset switch (rst_n) is asserted (0), the display pipeline correctly overrides standard output and renders the word "**n u L L**" across the first four 7-segment displays. The remaining two displays (t and t_c) remain completely off, confirming active-low reset functionality.